

Observation of Oriented Landau Levels and Helical Zero Modes in Berry Dipole Acoustic Crystals

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(Received 21 July 2024; revised 14 November 2024; accepted 18 February 2025; published 18 March 2025)

Landau levels play a fundamental role in condensed matter physics, as they offer critical insights into charged particles' behavior in magnetic fields. These unique characteristics have significant implications for understanding various quantum phenomena and materials' electronic properties, such as the renowned quantum Hall effect. Over the past two decades, substantial advancements have been achieved in Landau levels in systems with linear band crossing points, like Weyl points, which exhibit fascinating features like chiral zero modes. The Landau levels observed thus far are independent of the magnetic field's orientation. In this Letter, we experimentally demonstrate unconventional oriented Landau levels in acoustic Berry dipole systems, wherein by reversing the orientation of the pseudomagnetic field, the system exhibits distinct Landau spectra. Remarkably, we observe a new type of helical zero mode, whose existence critically depends on the magnetic field's orientation. This is in sharp contrast to the well-studied chiral zero modes in Weyl systems, whose propagation direction, but not existence, depends on the applied magnetic field's orientation. Our Letter opens up avenues for understanding the interaction of gauge fields with band singularities that go beyond Chern class, including Berry multipoles and even-dimensional monopoles, and offers new insights for exploring topological devices.

DOI: 10.1103/PhysRevLett.134.116604

Band crossing plays an important role in condensed matter theory, serving as the origin for various topological semimetals and insulators [1,2]. A well-known example is the Weyl point, which can be regarded as a source or sink of Berry curvature field in the momentum space, characterized by integer Chern numbers [3–8]. The presence of Berry monopole-antimonopole pairs gives rise to diverse exotic topological phenomena, including Fermi arc surface states [5,8,9], chiral anomaly [10–15], and quantum Hall effects [16]. Because of the isotropic distribution of the Berry curvature of Berry monopoles [Fig. 1(a)], the above topological phenomena are also isotropic. An instance of this isotropy can be seen when a magnetic field couples with a Weyl point, resulting in discrete Landau levels independent of the orientation of the magnetic field and a chiral zero mode unidirectionally propagating along

the direction of the magnetic field [17–27], as shown in Figs. 1(b) and 1(c), respectively.

Very recently, band crossing beyond the above paradigm was predicted and studied theoretically [28–30]. These band crossing points referred to as “Berry dipoles” carry a dipolar Berry curvature field resulting in a zero Chern number but nonzero Berry dipole moment and giving rise to exotic quantum geometric effects such as anomalous Hall conductivity and orbital magnetization [31,32]. Furthermore, the Berry dipole is considered a topological phase transition point from trivial insulators to 3D Hopf insulators [33–40], an emerging theoretical frame that goes beyond tenfold topological classification [41]. Interestingly, it was recently proposed that oriented Landau levels are formed when a Berry dipole is subject to an external magnetic field [31,32,42]. However, the experimental observation of this phenomenon remains elusive.

In this Letter, we experimentally realize oriented Landau levels and helical zero modes in Berry dipole semimetals constructed by engineering an inhomogeneous acoustic crystal to introduce pseudomagnetic fields. We demonstrate

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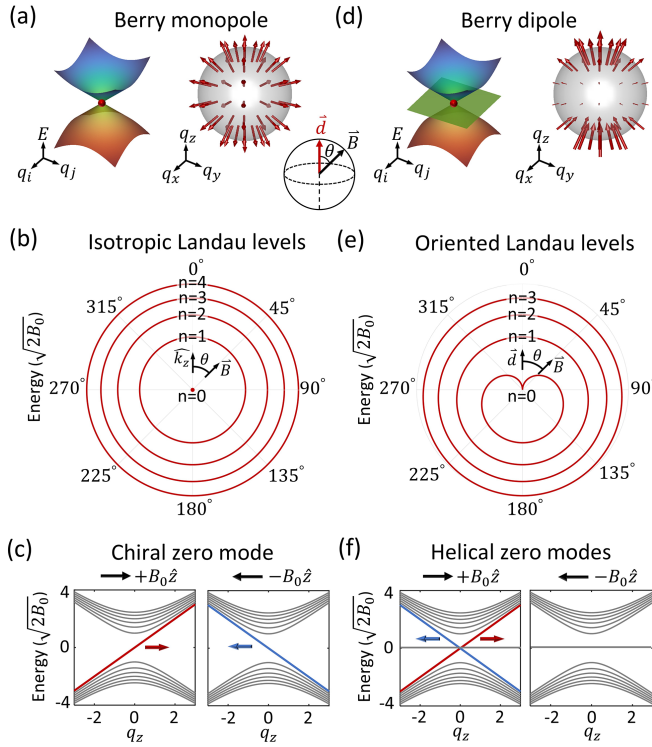


FIG. 1. (a),(d) The dispersion spectra (top panels) and Berry curvature distribution (bottom panels) in the Berry monopole (a) and dipole (d) systems, respectively. Inset: the polar angle of the magnetic field θ denotes the angle between the magnetic field $\vec{B}$ and the Berry dipole $\vec{d}$. (b),(e) The Landau level spectra versus θ in the Berry monopole (b) and dipole (e) systems. The dispersion spectra of the Berry monopole (c) and dipole (f) systems in the q_z direction when the magnetic field is along (top panels) and opposite (bottom panels) the z direction. Red and blue lines in (c) and (f) denote chiral zero modes and helical zero modes, respectively.

dramatically different Landau level spectra when the pseudomagnetic field is parallel or antiparallel to the Berry dipole moment. Specifically, when the pseudomagnetic field is parallel to the Berry dipole moment, there exists a pair of helical zero modes propagating, respectively, along and opposite the pseudomagnetic field. Remarkably, the helical zero modes vanish, and Landau levels are gapped when the pseudomagnetic field is reversed. Our Letter offers a platform to study topological matter beyond Chern class, such as Berry multipoles and four-dimensional tensor monopoles.

We start from a minimal three-band continuum model of a Berry dipole [28]:

$$H_{\text{BD}} = q_x \lambda_4 + q_y \lambda_5 + q_z \lambda_7$$

$$= \begin{pmatrix} 0 & 0 & q_x - iq_y \\ 0 & 0 & -iq_z \\ q_x + iq_y & iq_z & 0 \end{pmatrix}, \quad (1)$$

where $\vec{q} = (q_x, q_y, q_z)$ is the 3D momentum vector deviated from the Berry dipole, and λ_i represents the 3×3 Gell-Mann matrices [43]. Figure 1(d) shows the dispersion structure and distribution of Berry curvature near the triply degenerated nodal point at $\vec{q} = 0$. The band structure exhibits an isotropic configuration consisting of a flat band and two linear crossing bands with Weyl-like dispersions. The distribution of Berry curvature near the nodal point is characterized by a dipolar field distribution of $\Omega = \kappa(\vec{d} \cdot \vec{q})\vec{q}/(2|q|^4)$, where $\kappa = 1$ and $\vec{d} = (0, 0, 1)$ represent the strength and orientation of the Berry dipole moment [44].

Next, considering a strong magnetic field applied to the Berry dipole system, the Landau levels can be exactly solved as

$$E = \pm \sqrt{2B_0 \left(n + \frac{1 - \cos \theta}{2} \right)} + q_0^2, \quad (2)$$

where B_0 is the strength of the magnetic field, θ is the angle between the magnetic field $\vec{B}$ and Berry dipole moment $\vec{d}$, $q_0 = \vec{B} \cdot \vec{q}/|\vec{B}|$ is the projection of the momentum along the direction of magnetic field, and $n \in \{0, 1, 2, \dots\}$ [44]. As shown in Fig. 1(e), the Landau level spectrum against θ has an anisotropic form with $q_0 = 0$, which is different from the circular shape of Landau levels in a Berry monopole system [Fig. 1(b)]. Remarkably, the zeroth Landau levels of the Berry dipole exhibit a heart-shaped contour with a cusp at $\theta = 0$. We focus on two extreme cases ($\theta = 0$ and π) to reveal the orientation-dependent properties. When $\theta = 0$, a pair of gapless helical zero modes (red and blue lines) connect the bottom and top bulk bands with the group velocities in both positive and negative z directions, as shown in the top panel of Fig. 1(f). In contrast, when the pseudomagnetic field is reversed ($\theta = \pi$), the helical zero modes vanish, and a band gap ($2\sqrt{2B_0}$) appears between two zeroth Landau levels, as shown in the bottom panel of Fig. 1(f). This behavior is akin to a field-effect transistor, where the “ON” and “OFF” states are controlled by the orientation of the applied magnetic field. It is worth noting that the oriented Landau levels effect, whose spectrum depends on the geometry of the Berry curvature distribution, is a pure quantum geometric effect and cannot be revealed by the semiclassical quantization method [44].

To realize the oriented Landau levels in realistic physical systems, we propose a lattice model possessing Berry dipoles and introduce a pseudomagnetic field by engineering the inhomogeneous couplings. The lattice model and corresponding energy spectrum between high-symmetry points are shown in Figs. 2(a) and 2(b) when $t_2 = t_1$. This configuration features Berry dipoles (depicted as yellow arrows) located at the K and K' valleys, oriented toward the positive and negative k_z directions. By adjusting the

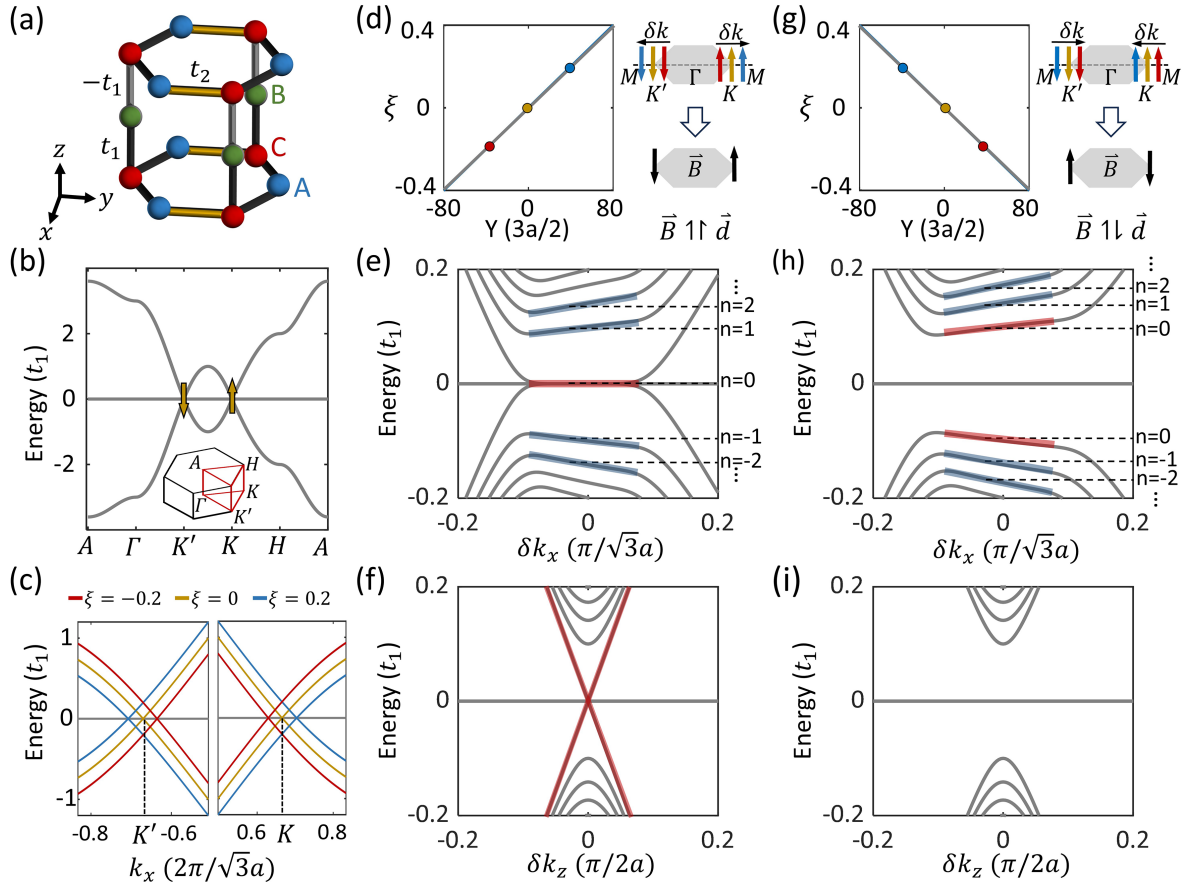


FIG. 2. (a) The tight-binding model for the Berry dipole semimetal. The black (gray) and yellow tubes represent the strength of nearest-neighbor coupling $\pm t_1$ and t_2 . $t_2 = (1 + \xi)t_1$. The nearest-neighbor distance between sublattices is a . (b) The energy spectrum with $\xi = 0$ in the $A - \Gamma - K' - K - H - A$ line. The yellow arrows denote Berry dipoles at the K and K' valleys. (c) The band spectra with $\xi = -0.2, 0, 0.2$. (d),(g) The relation of the coupling coefficient ξ to the y position, which is an increasing and decreasing function in (d) and (g), respectively. The insets show the trajectory of Berry dipoles as the y position increases and corresponding pseudomagnetic fields $[\vec{B} \uparrow \uparrow \vec{d}]$ in (d) and $[\vec{B} \downarrow \downarrow \vec{d}]$ in (g). (e),(f) Energy spectra in the k_x and k_z direction for (d), respectively. (h),(i) Energy spectra in the k_x and k_z direction for (g), respectively. $(\delta k_x, \delta k_z) = (k_x - k_{x0}, k_z - k_{z0})$, where (k_{x0}, k_{z0}) is the location of the Berry dipole when $\xi = 0$.

coupling $t_2 = (1 + \xi)t_1$, where ξ is a dimensionless parameter, the Berry dipole at the K (K') valley can be displaced in the k_x direction, as shown in Fig. 2(c). Through spatially modulating t_2 as $\xi = B_0(3a/2)y$, one can approximately define the gauge field $A_x = \mp B_0 y$ at the $K(K')$ valley. Consequently, the pseudomagnetic field can be obtained by $\vec{B} = \nabla \times \vec{A} = \pm B_0 \hat{z}$ at the $K(K')$ valley. When B_0 is positive, the Berry dipoles shift in the $\Gamma - K(K') - M$ line as y increases, as illustrated in Fig. 2(d), leading to a pseudomagnetic field parallel to the Berry dipoles ($\vec{B} \uparrow \uparrow \vec{d}$) at both K and K' valleys. The system hosts gapless zeroth Landau levels at K and K' valleys, behaving as zero-energy flat bands in the k_x direction and helical zero modes in the k_z direction, as shown in Figs. 2(e) and 2(f). Conversely, for negative B_0 , the Berry dipoles shift in the $M - K(K') - \Gamma$ line as y increases, as shown in Fig. 2(g), resulting in a pseudomagnetic field antiparallel to the Berry dipoles

($\vec{B} \downarrow \downarrow \vec{d}$) at both K and K' valleys. Figures 2(h) and 2(i) show the energy spectra of Landau levels in the k_x and k_z direction near the K' valley, respectively. The zeroth Landau levels are gapped out, and no propagation states are allowed in the band gap. Moreover, one can realize any angle between $\vec{B}$ and $\vec{d}$ by spatially modulating t_1 in different directions [44].

Inspired by the lattice model, we design acoustic crystals to demonstrate the oriented Landau level effect in the experiment. As shown in Fig. 3(a), the acoustic unit cell consists of three acoustic resonators (blue blocks) corresponding to three sublattices “A,” “B,” and “C” in Fig. 2(a), and the channels between them (gray and yellow blocks) provide the nearest-neighbor couplings. This configuration can realize the Hamiltonian of the hexagonal lattice in an easily processable tetragonal-like crystal [44]. Here, we consider the dipole modes in acoustic resonators,

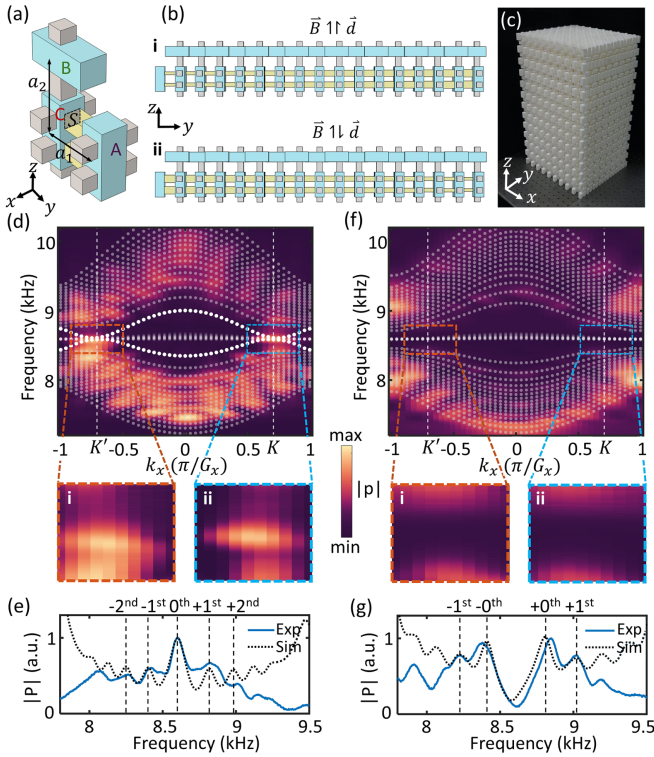


FIG. 3. (a) Unit cell of the acoustic crystal. a_1 (a_2) is the nearest-neighbor distance between the A and C (B and C) sublattices. (b) Two configurations of inhomogeneous acoustic supercells corresponding to $\vec{B} \parallel \vec{d}$ and $\vec{B} \perp \vec{d}$, respectively. (c) Photographs of two experimental samples with the same appearance but different configurations in (b). (d),(f) Measured (colored areas) and simulated (gray and white dots) Landau levels in the k_x direction with $k_z = 0$ in samples 1 ($\vec{B} \parallel \vec{d}$) and 2 ($\vec{B} \perp \vec{d}$), respectively. $G_x = 2a_1$. The insets (i) and (ii) are enlarged views near the K' and K valleys. (e),(g) Measured (blue solid lines) and simulated (black dashed lines) acoustic pressure spectra at the K valley for samples 1 and 2, respectively.

whose simulated band structure matches well with that of the tight-binding model [44]. We control the area of the cross section of yellow channels $S = (1 + \xi)S_0$, which has an approximately linear relationship with the strength of intracell coupling between the A and C resonators but does not affect the frequency of the Berry dipoles [44]. More detailed structure parameters are presented in Fig. S5 of Supplemental Material [44].

Then, we construct a supercell containing 16 units in the y direction and spatially modulate ξ in the range of -0.75 to 0.75 , which introduces a strong pseudomagnetic field in the z direction with the strength of $|\vec{B}| = 0.067a_1^{-2}$. As illustrated in Fig. 3(b), when y increases, ξ is linearly increased (decreased) to introduce a pseudomagnetic field parallel (antiparallel) to the Berry dipoles [44]. In addition, one can also modulate inhomogeneous ξ in different directions to realize the pseudomagnetic field in different orientations, such as the case of $\theta = \pi/2$ [44].

To experimentally observe the oriented Landau levels, we fabricated two samples containing $18 \times 16 \times 12$ acoustic units with a 3D-printing technique [Fig. 3(c)]. Samples 1 and 2 have the same configurations except for the different variations of inhomogeneous channels in the y direction as shown [Fig. 3(b)] corresponding to $\vec{B} \parallel \vec{d}$ and $\vec{B} \perp \vec{d}$, respectively. To distinguish the zeroth Landau levels from the middle flat bulk band, only the C sublattices are excited and measured, whose sound pressure field has no contribution from the middle band [44]. A sound source was placed in the center of the bottom surface of the sample, and a microphone was inserted into the sample to scan the sound pressure field distribution on the x - z plane. After a 2D Fourier transformation, we obtained the 2D projected dispersion spectra in the k_x - k_z plane [44]. Figures 3(d) and 3(f) show the measured dispersion spectra in the k_x direction with fixed $k_z = 0$ for samples 1 and 2, respectively, together with the numerical dispersion spectra. Enlarged views of the dispersion near the K' and K valleys show the presence of gapless flat zeroth Landau levels around the frequency of 8600 Hz in sample 1, while a band gap between two zeroth modes in sample 2.

For a more quantitative characterization of the Landau levels, we further present the spectra of samples 1 and 2 at the K valley in Figs. 3(e) and 3(g), respectively. Each spectrum exhibits several peaks corresponding to the locations of the Landau levels. For sample 1, there is a pronounced peak right in the middle of the spectrum corresponding to the gapless zeroth modes. In contrast, in sample 2 the two zeroth modes are separated, forming a gap in the middle. The locations of the Landau levels in the measured spectra agree well with those of the simulation, providing direct evidence of the oriented Landau levels.

We next focus on the oriented helical zero modes and their propagating properties in the direction of the pseudomagnetic field in samples 1 and 2. For sample 1, a pair of gapless helical zero modes is present in each valley with negative and positive group velocities, as shown in Fig. 4(a). The sound pressure field distributions in the y - z plane are also measured at the exciting frequency of 8500 Hz, as shown in the left panels of Figs. 4(a) and 4(c), corresponding to sound sources placed on the top and bottom surfaces, respectively. It is seen that the sound wave propagates along the z direction and follows a Gaussian distribution in the y direction, agreeing well with the simulated sound pressure field distributions in the right panels of Figs. 4(b) and 4(c).

The measured dispersion for sample 2 is shown in Fig. 4(d). Because of the gap between the two Landau zeroth modes, the propagation of sound within this frequency range is prohibited. Under the excitation at 8500 Hz, the measured sound pressure is localized at the source position and exponentially decays into the bulk, as shown in the left panels of Figs. 4(e) and 4(f). The measurement results

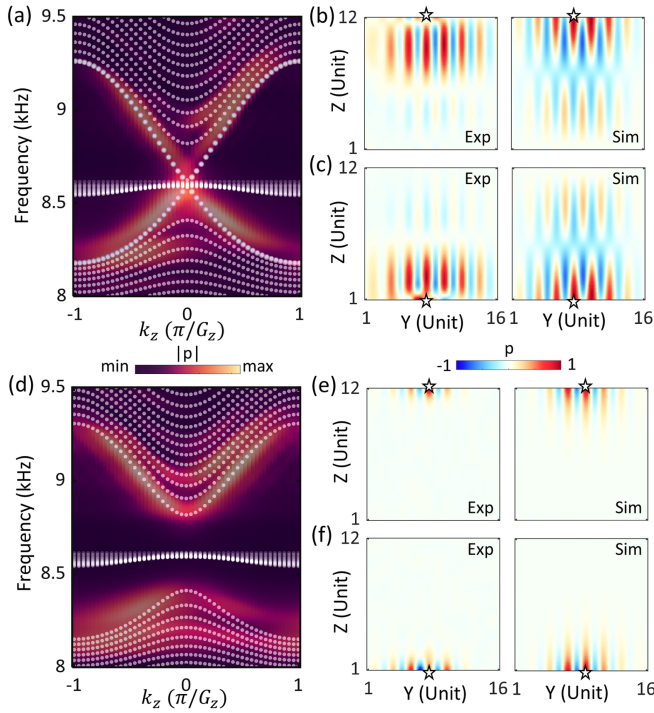


FIG. 4. (a) Measured dispersion spectra of helical zero modes in the K valley in sample 1 ($\vec{B} \parallel \vec{d}$). $G_z = 2a_2$. (b),(c) Sound pressure distributions of down-going and up-going zeroth modes at the frequency of 8500 Hz, respectively. Left and right panels represent experimental and simulated results. The sources (pentagrams) are located on the top and bottom surfaces, respectively. (d) Measured dispersion spectra of gapped Landau levels in the K valley in sample 2 ($\vec{B} \parallel \vec{d}$). (e),(f) Sound pressure distributions at the frequency of 8500 Hz in experiment (left panels) and simulation (right panels).

agree well with the simulated sound pressure field distributions in the right panels of Figs. 4(e) and 4(f).

To sum up, we have experimentally demonstrated oriented Landau levels in the 3D Berry dipole system via the introduction of pseudomagnetic fields in an acoustic system. Our Letter opens the door to exploring the effect of gauge field on band crossing points beyond conventional Chern-class crossing points. Meanwhile, our acoustic design provides a platform for studying exotic topological phenomena related to Berry dipoles, such as 3D Hopf insulators and even tensor monopoles in synthetic 4D acoustic lattice. The distinct Landau levels exhibited under different orientations of pseudomagnetic field may offer novel insights for the design of optical devices.

Acknowledgments—We acknowledge the support of New Cornerstone Science Foundation, the Research Grants Council of Hong Kong (STG3/E-704/23-N, AoE/P-502/20, 17309021), the National Natural Science Foundation of China (12274031), National Key R&D Program of China (No. 2022YFA1404500), Beijing Institute of Technology Research Fund Program for Teli

Young Fellows, Beijing Institute of Technology Innovative Talents Science and Technology Funding Special Plan (2022CX01006), Guangdong Provincial Quantum Science Strategic Initiative (GDZX2204004, GDZX2304001), Guangdong Basic and Applied Basic Research Foundation (No. 2021B1515020086).

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Supplementary Material for
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Section 1. Berry curvature of Berry dipole

In this section, we provide detailed calculations of Berry curvature distributions of the three-band Berry dipole, whose Hamiltonian can be written as

$$H_{BD} = \eta q_x \lambda_4 + q_y \lambda_5 + q_z \lambda_7$$

$$= \begin{pmatrix} 0 & 0 & \eta q_x - i q_y \\ 0 & 0 & -i q_z \\ \eta q_x + i q_y & i q_z & 0 \end{pmatrix}, \quad (S1)$$

where $\eta = \pm 1$ represent the valley index, and λ_i are Gell-Mann matrices [1]. The Hamiltonian is protected by chiral symmetry $\{H_{BD}, U\} = 0$, where $U = \text{diag}(1, 1, -1)$. To simplify calculations, we rewrite the Hamiltonian in spherical coordinate system, which can be written as

$$H_{BD} = \begin{pmatrix} 0 & 0 & \eta r \sin \theta e^{-i \eta \phi} \\ 0 & 0 & -i r \cos \theta \\ \eta r \sin \theta e^{i \eta \phi} & i r \cos \theta & 0 \end{pmatrix} \quad (S2)$$

where the momenta $\vec{q} = (q_x, q_y, q_z)$ is replaced by (r, θ, ϕ) . The eigenvalues of the Hamiltonian are $E = 0, \pm r$. The eigenvector of the lowest band can be written as $\psi = (\eta \sin \theta e^{-i \eta \phi}, -i \cos \theta, -1)^T / \sqrt{2}$. Then, we calculate the Berry connection and Berry curvature, which can be written as

$$\vec{A} = \langle \psi | i \partial_j | \psi \rangle = \left(0, 0, \frac{\eta \sin \theta}{2r} \right), \quad (S3)$$

$$\vec{\Omega} = \nabla \times \vec{A} = (\partial_i A_j - \partial_j A_i) \hat{e}_k = \frac{\eta \cos \theta}{r^2} \hat{r}. \quad (S4)$$

Comparing Eq. (S4) with the standard dipolar field $\Omega = \kappa (\vec{d} \cdot \vec{q}) \vec{q} / |q|^4$, we can readily see that the strength of Berry dipole moment $\kappa = 1$ and the orientation vector $\vec{d} = (0, 0, \eta)$, a unit vector in the k_z direction. Fig. 1(d) shows the case of $\eta = +1$ and the Berry dipole moment points in the positive k_z direction.

Section 2. Landau levels in Berry monopole and Berry dipole systems

In this section, we will compare similarities and differences between the dispersion spectra of Landau levels in Berry monopole and Berry dipole systems, as shown in Fig. 1(a). For simplicity, we focus on the Weyl point as a typical Berry monopole with the Hamiltonian of $H_{BM} = q_x\sigma_1 + q_y\sigma_2 + q_z\sigma_3$, and the minimal three-band model of Berry dipole protected by chiral symmetry with Hamiltonian of $H_{BD} = \eta q_x\lambda_4 + q_y\lambda_5 + q_z\lambda_7$. Due to the axial rotation symmetry of Berry curvature in both Berry monopole and dipole systems, we consider a vector gauge potential $\vec{A} = B_0 y(-\cos\theta, 0, \sin\theta)$, corresponding to magnetic field $\vec{B} = B_0(\sin\theta, 0, \cos\theta)$, without losing of generality. Here, B_0 and θ represent the strength and polar angle of magnetic field, respectively. In the presence of the strong magnetic field, we replace the canonical momentum q by the gauge-invariant kinetic momentum $q_i \rightarrow \Pi_i \equiv -i\partial_i + A_i$. To be more specific, three momentum terms can be written as

$$\begin{aligned}\Pi_x &= -i\partial_x - B_0 y \cos\theta, \\ \Pi_y &= -i\partial_y, \\ \Pi_z &= -i\partial_z + B_0 y \sin\theta.\end{aligned}\tag{S5}$$

Using the canonical commutation relations $[x_j, q_k] = i\delta_{jk}$, we can write down the commutation relations between the gauge-invariant kinetic momenta as

$$\begin{aligned}[\Pi_x, \Pi_y] &= -iB_0 \cos\theta, \\ [\Pi_y, \Pi_z] &= -iB_0 \sin\theta, \\ [\Pi_z, \Pi_x] &= 0.\end{aligned}\tag{S6}$$

Then, we define creation and annihilation operators with the form of

$$\begin{aligned}\hat{a}^\dagger &= \frac{1}{\sqrt{2B_0}}(\Pi_y - i\cos\theta\Pi_x + i\sin\theta\Pi_z), \\ \hat{a} &= \frac{1}{\sqrt{2B_0}}(\Pi_y + i\cos\theta\Pi_x - i\sin\theta\Pi_z),\end{aligned}\tag{S7}$$

which satisfy the canonical commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$. Moreover, $\Pi_x^2 + \Pi_y^2 + \Pi_z^2 = B_0(2\hat{a}^\dagger\hat{a} + 1) + q_0^2$, where $q_0 = \Pi \cdot B/B_0 = \sin\theta\Pi_x + \cos\theta\Pi_z$ represents the

component of momenta in the direction of the magnetic field. Furthermore, we can rewrite the momenta by creation and annihilation operators as

$$\begin{aligned}\Pi_x &= i \sqrt{\frac{B_0}{2}} \cos\theta (\hat{a}^\dagger - \hat{a}) + q_0 \sin\theta, \\ \Pi_y &= \sqrt{\frac{B_0}{2}} (\hat{a}^\dagger + \hat{a}), \\ \Pi_z &= i \sqrt{\frac{B_0}{2}} \sin(\hat{a} - \hat{a}^\dagger) + q_0 \cos\theta.\end{aligned}\tag{S8}$$

2.1 Isotropic Landau levels in Berry monopole system

For the Weyl point (Berry monopole), three Pauli matrices satisfy a Clifford algebra with the form of $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$, and the canonical commutation relation is $[\sigma_j, \sigma_k] = 2i\sigma_{jk}$. Then, we square the Hamiltonian of Weyl point,

$$\begin{aligned}H_{BM}^2 &= \sum_i \Pi_i^2 + i[\Pi_x, \Pi_y]\sigma_z + i[\Pi_y, \Pi_z]\sigma_x \\ &= \Sigma^2 + B_0 \cos\theta \sigma_z + B_0 \sin\theta \sigma_x,\end{aligned}\tag{S9}$$

where $\Sigma^2 = B_0(2\hat{a}^\dagger \hat{a} + 1) + q_0^2$. Solving the eigen-equation $(\Sigma^2 - \varepsilon^2)\psi = -(B_0 \cos\theta \sigma_z + B_0 \sin\theta \sigma_x)\psi$, we obtain the following eigenvalues $B_0(2\hat{a}^\dagger \hat{a} + 1) + q_0^2 - \varepsilon^2 = \pm B_0$. Through defining a number operator $n = \hat{a}^\dagger \hat{a}$, the energy spectrum of Landau levels of Weyl point can be exactly solved as

$$\varepsilon = \text{sgn}(n) \sqrt{B_0(2|n|) + q_0^2},\tag{S10}$$

where $n \in \{0, \pm 1, \pm 2, \dots\}$. When $q_0 = 0$, Landau levels are doubly degenerated, which is expected for the zeroth Landau level with zero energy. The zeroth mode breaks chiral symmetry and propagates unidirectionally along the magnetic field. Such phenomenon is known as ‘‘Chiral anomaly’’, which originates from the chirality of the Weyl point [2–4]. Furthermore, the Landau level spectrum of Weyl point is isotropic and independent on the polar angle of magnetic field θ , as shown in Fig. 1(b) in the main text.

2.2 Oriented Landau levels in Berry dipole system

For the three-band Berry dipole, the Hamiltonian in Eq. (S1) is protected by the chiral symmetry $\{H_{BD}, U\} = 0$, where $U = \text{diag}(1, 1, -1)$. Therefore, there is no

chiral anomaly in the chiral symmetric Berry dipole system, which is distinct from the Weyl point. The Clifford algebra of Gell-Mann matrices is more complex, so we directly solve the Landau levels from the eigen-equation. We set the eigenstates are $(\psi_1, \psi_2, \psi_3)^T$, and the eigen-equation of Berry dipole can be written as

$$\begin{pmatrix} 0 & 0 & \eta\Pi_x - i\Pi_y \\ 0 & 0 & -i\Pi_z \\ \eta\Pi_x + i\Pi_y & i\Pi_z & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix} = \varepsilon \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix}. \quad (S11)$$

Then, we define two-component spinors as $Q = (\eta\Pi_x + i\Pi_y, i\Pi_z)^T$ and $\psi_{12} = (\psi_1, \psi_2)^T$. The eigen-equation can be rewritten as

$$\begin{aligned} Q^\dagger \psi_{12} &= \varepsilon \psi_3, \\ \varepsilon \psi_{12} &= Q \psi_3. \end{aligned} \quad (S12)$$

We can write down the eigen-equations corresponding to ψ_{12} and ψ_3 as

$$\begin{aligned} Q Q^\dagger \psi_{12} &= \varepsilon^2 \psi_{12}, \\ Q^\dagger Q \psi_3 &= \varepsilon^2 \psi_3. \end{aligned} \quad (S13)$$

We focus on the eigen-equation of ψ_3 . $Q^\dagger Q = \Sigma^2 - i\eta[\Pi_x, \Pi_y] = B_0(2\hat{a}^\dagger \hat{a} + 1 - \eta \cos\theta) + q_0^2$. Therefore, the expression of Landau levels in Berry dipole system is characterized as

$$\varepsilon = \pm \sqrt{2B_0 \left(n + \frac{1 - \eta \cos\theta}{2} \right) + q_0^2}. \quad (S14)$$

The orientation-dependent term $\eta \cos\theta = \cos\beta$, where θ is the polar angle of Berry dipole moment and β is the angle between Berry dipole moment and magnetic field. In the main text, $\eta = +$, which means $\theta = \beta$. Furthermore, due to chiral symmetry of the whole system, Landau level modes always appear in a pair with opposite energies, even the zeroth modes. As we can see the case of $\theta = 0$ in Fig. 4, each valley hosts a pair of gapless zeroth modes propagating on both $+z$ and $-z$ directions.

Furthermore, we solve the eigenstates of oriented Landau levels in the Berry dipole system. From the second line of Eq. (S13), it is easy to see $\psi_3 \sim |n\rangle$, where $|n\rangle$ is the eigenstate of quantum harmonic oscillator. Combining with Eq. (S8) and (S11), we can get the forms of ψ_1 and ψ_2 , which can be written as

$$\begin{aligned}
\varepsilon\psi_1 &\sim i\sqrt{\frac{B_0(n+1)}{2}}(\eta\cos\theta - 1)|n+1\rangle + \eta q_0\sin\theta|n\rangle - i\sqrt{\frac{B_0n}{2}}(\eta\cos\theta + 1)|n-1\rangle, \\
\varepsilon\psi_2 &\sim -\sqrt{\frac{B_0(n+1)}{2}}\sin\theta|n+1\rangle - i\eta q_0\cos\theta|n\rangle + \sqrt{\frac{B_0n}{2}}\sin\theta|n-1\rangle. \quad (S15)
\end{aligned}$$

Remarkably, when $\theta = 0$ ($\vec{B}$ parallel to $\vec{d}$), the zeroth Landau levels serve as a pair of gapless zero modes in the k_z direction with the energy $\varepsilon = \pm q_z$ and the eigenstates $\boldsymbol{\psi}_{up} = \frac{(\mathbf{0}, i, \mathbf{1})^\dagger q_z}{\sqrt{2}|q_z|} |\mathbf{0}\rangle$ for the upward-propagating mode ($\varepsilon = +q_z$) and $\boldsymbol{\psi}_{down} = \frac{(\mathbf{0}, i, -\mathbf{1})^\dagger q_z}{\sqrt{2}|q_z|} |\mathbf{0}\rangle$ for the downward-propagating mode ($\varepsilon = -q_z$). The upward- (downward-) propagating mode $\psi_{up(down)}$ is very similar to the spin-up (spin-down) state in the QSHE, which can be regarded as helical zero modes. By contrast, when $\theta \neq 0$, the zeroth Landau levels are gapped out and the spin is ill-defined, leading to the vanishment of the helical zero modes.

Section 3. Semiclassical quantization method for Berry dipole system

In this section, we calculate the Landau level spectrum of the Berry dipole system through a semiclassical way. We decompose the momenta into two parts, one is q_0 in the $\hat{B}$ direction, and the other is $q_\perp$ in the plane perpendicular to the magnetic field. The eigenvalues are $E = \pm\sqrt{q_0^2 + q_\perp^2}$ and 0. Here we consider the lowest band with the energy of $-\sqrt{q_0^2 + q_\perp^2}$. With a fixed q_0 , the momentum component perpendicular to the magnetic field $q_\perp^2 = E^2 - q_0^2$ represents a series of closed orbits in the plane perpendicular to the magnetic field. According to Onsager quantization, the area of these orbits should be quantized as

$$S = \pi q_\perp^2 = 2\pi B(n + \gamma), \quad (S16)$$

where $\gamma = 1/2$ for circular orbits and $n \in \{0, 1, 2 \dots\}$. Therefore, the expression of quantized Landau levels can be written as

$$E = \pm \sqrt{2B \left(n + \frac{1}{2} \right) + q_0^2}. \quad (S17)$$

As we can see, the Landau levels calculated from the semiclassical method cannot reveal the geometric property of the Berry dipole system. This is because that above calculations come from the isotropic dispersion spectrum of the Berry dipole instead of dipolar Berry curvature field, which cause a lack of the geometry-dependent term in the Landau level spectra.

Section 4. The pseudomagnetic field induced by inhomogeneous couplings

First, the global Hamiltonian of the lattice model without inhomogeneous couplings can be written as

$$H = \begin{pmatrix} 0 & 0 & t_1 + 2t_2 \cos \frac{\sqrt{3}ak_x}{2} e^{i\frac{3aky}{2}} \\ 0 & 0 & t_1 - t_2 e^{i2ak_z} \\ t_1 + 2t_2 \cos \frac{\sqrt{3}ak_x}{2} e^{-i\frac{3aky}{2}} & t_1 - t_2 e^{-i2ak_z} & 0 \end{pmatrix}. \quad (S18)$$

When $t_2 = t_1$, A and C sites form a graphene-like lattice in the x-y plane, while B and C sites form a chain with straggled opposite couplings in the z direction. There are two triply degenerated nodal points located at the K and K' valleys, whose effective Hamiltonian can be written as

$$H_{eff} = t_1 \left\{ \eta \frac{3a}{2} \delta k_x \lambda_4 + \frac{3a}{2} \delta k_y \lambda_5 + 2a \delta k_z \lambda_7 \right\} \\ = t_1 \begin{pmatrix} 0 & 0 & \eta \frac{3a}{2} \delta k_x - i \frac{3a}{2} \delta k_y \\ 0 & 0 & -i2a \delta k_z \\ \eta \frac{3a}{2} \delta k_x + i \frac{3a}{2} \delta k_y & i2a \delta k_z & 0 \end{pmatrix}, \quad (S19)$$

where $\eta = \pm$ at the K (K') valley. Therefore, the triply degenerated nodal point at the K (K') valley can be considered as the Berry dipole pointing in the positive (negative) k_z direction.

Next, we slightly change $t_2 = (1 + \xi)t_1$, where $|\xi| \ll 1$. The effective Hamiltonian near the K (K') valley can be written as

$$H_{eff} = t_1 \left\{ \left(\eta \frac{3a}{2} \delta k_x - \xi \right) \lambda_4 + \frac{3a}{2} \delta k_y \lambda_5 + 2a \delta k_z \lambda_7 \right\}, \quad (S20)$$

which shifts the Berry dipole in the positive (negative) k_x direction at the K (K') valley with ξ increasing, as shown in Fig. 2(c). Here, we can replace the canonical momentum q_x by the gauge-invariant kinetic momentum as $q_x \rightarrow \Pi_x = q_x + eA_x$, where $e = 1$ for simplification and $A_x = -\frac{\eta\xi}{3a/2}$.

Finally, we spatially modulate ξ as a linear function to the position y , which effectively introduces a magnetic vector potential $A_x = -\eta B y$, where B is the slope of the $\xi(y)$ function in Fig. 2(d, g). The pseudomagnetic field $\vec{B} = \nabla \times \vec{A} = -\partial_y A_x \hat{z} = \eta B \hat{z}$. As you can see, the direction of pseudomagnetic field depends on the slope of $\xi(y)$ function and the valley index η . When B is positive (Fig. 2(d)), the pseudomagnetic

field is along the positive (negative) z direction at the K (K') valley, corresponding to the pseudomagnetic field parallel to the Berry dipoles at both K and K' valleys. In the contrast, when B is negative (Fig. 2(g)), the pseudomagnetic field is antiparallel to the Berry dipoles at both K and K' valleys.

Section 5. Landau levels for different θ

We can realize the pseudomagnetic field in any θ direction in our inhomogeneous model. By spatially modulating the strength of t_2 coupling as $\xi = B(y\cos\theta - z\sin\theta)$, we can effectively introduce a gauge field $A_x = -\eta B(y\cos\theta - z\sin\theta)$, corresponding to the pseudomagnetic field $\vec{B} = \nabla \times \vec{A} = -\partial_y A_x \hat{z} + \partial_z A_x \hat{y} = \eta B(\sin\theta \hat{y} + \cos\theta \hat{z})$. The angle between the pseudomagnetic field and the Berry dipole is θ at both K and K' valleys. For example, to realize the pseudomagnetic field perpendicular to the Berry dipole moment, we can construct a supercell containing 161 units in the z direction, and linearly control the inhomogeneous coupling t_2 from $1.4t_1$ to $0.6t_1$ when y increases. Such spatial arrangement corresponds to a pseudomagnetic potential $\vec{A} = (Bz, 0, 0)$, which introduces the pseudomagnetic fields perpendicular to Berry dipole moment in both K and K' valleys (the case of $\theta = 0$). The calculated dispersion spectra in the k_x direction with $k_y = 0$ and in the k_y direction with $k_x = -4\pi/(3\sqrt{3}a)$ are shown in Fig. S1(a), respectively. There is an energy gap between the zeroth Landau levels, whose value is between the case of $\theta = 0$ and π [Fig. S1(b, c)].

Section 6. The Hamiltonian of the tetragonal-like lattice

To facilitate sample fabrication and measurement, we deform the hexagonal lattice in Fig. S2(a) to a tetragonal-like lattice in Fig. S2(c). The Hamiltonian of this tetragonal-like lattice can be written as

$$H = \begin{pmatrix} 0 & 0 & t_1 + 2t_2 \cos(ak_x)e^{iak_y} \\ 0 & 0 & t_1 - t_1 e^{i2ak_z} \\ t_1 + 2t_2 \cos(ak_x)e^{-iak_y} & t_1 - t_1 e^{-i2ak_z} & 0 \end{pmatrix}. \quad (S21)$$

Compared with the Eq. (S18), the only difference is the lattice constant in the x-y plane.

As shown in Fig. S2(b, d), the x-y lattice constant in the hexagonal lattice is $\sqrt{3}a$ and

$\vec{a}_{1,2} = (\sqrt{3}a/2, \pm 3a/2, 0)$, while the x-y lattice constant in the tetragonal-like lattice is

$\sqrt{2}a$ and $\vec{a}_{1,2} = (a, \pm a, 0)$. These differences only make a deformation of the first

Brillouin zone but do not change the tight-binding Hamiltonian.

Section 7. The strength of coupling between acoustic resonators

In this section, we discuss the strength of coupling between acoustic resonators. We consider two resonators connected by yellow coupling channels, as shown in Fig. S4(a). This system can be simply characterized by a 2×2 Hamiltonian as

$$H = \begin{pmatrix} f & \kappa \\ \kappa & f \end{pmatrix}, \quad (S22)$$

where f and κ represent the eigenfrequency of each resonator and the coupling strength, respectively. The gap between two eigenfrequencies in this two resonators system reveals the coupling strength. We depict the simulated coupling strength when changing the dimensional parameter ξ from 0.25 to 1.75 in Fig. S4(b). As we can see, the coupling strength linearly increases from 200 Hz to 1100 Hz when the area of cross section of coupling channel increases. We also plot the center frequencies of this system against ξ in Fig. S4(c), whose variation range is within 100 Hz.

Section 8. Realization of the case of $\theta = \pi/2$ in the acoustic lattice

To realize the pseudomagnetic field perpendicular to the Berry dipole moment corresponding to the case of $\theta = \pi/2$, we design an acoustic supercell containing 16 cells in the z direction and spatially modulate ξ in the range of 0.25 to 1.75. As shown in Fig. S6(a), we linearly decrease the area of the cross-section of yellow coupling channels when y increases, leading to a pseudomagnetic vector in the form $\vec{A} = (B_0 z, 0, 0)$, corresponding to a pseudomagnetic field $\vec{B} = (0, B_0, 0)$, where the strength of $|B| = 0.067 a_1^{-2}$. We plot the simulated dispersion spectra in the k_x direction with $k_y = 0$ and in the k_y direction with $k_x = 2\pi/(3a_1)$ in Fig. S6(b, c), respectively. At the K (K') valley (black dash line in Fig. S6(b)), there is a frequency gap of about 210 Hz between gapped zeroth Landau levels (red lines). Moreover, higher-order Landau levels are shown in blue and green lines.

Section 9. Sublattice effect in the Berry dipole semimetal

We rewrite the Hamiltonian of the Berry dipole semimetal to a general form as

$$H = \begin{pmatrix} 0 & 0 & d_1 - id_2 \\ 0 & 0 & d_3 - id_4 \\ d_1 + id_2 & d_3 + id_4 & 0 \end{pmatrix}, \quad (S23)$$

where d_{1-4} indicate components of the Hamiltonian of the Berry dipole semimetal in four Gell-Mann matrices. The eigenvalues of Eq. (S20) are $E = 0$ and $\pm \sqrt{d_1^2 + d_2^2 + d_3^2 + d_4^2}$, which form a flat band and two linear crossing bands. For the flat band with $E = 0$, the eigenvector is $\{(d_3 + id_4)/|d|, -(d_1 + id_2)/|d|, 0\}$, where $|d| = \sqrt{d_1^2 + d_2^2 + d_3^2 + d_4^2}$. As we can see, the modes of the flat band do not distribute in C sublattices. For the dispersive bands with $E = \pm|d|$, the eigenvectors are $\{(d_1 - id_2)/|d|, (d_3 - id_4)/|d|, \pm 1\}/\sqrt{2}$, respectively. Therefore, to distinguish the gapless zeroth modes in the case of $\theta = 0$ from the flat band, we can pump and probe only in the C sublattices.

Section 10. Experimental setup and measurements

The experimental setup is schematically shown in Fig. S7. A micro-speaker with diameter of 5 mm was placed at a C sublattice at the bottom or top of the sample as a sound source, and a microphone (Knowles 3019) with the size of $3\text{ mm} \times 1.9\text{ mm} \times 0.9\text{ mm}$ was inserted into the sample as sound signal detector. The micro-speaker worked in the form of continuous excitation by scanning frequency from 7.2 kHz to 10.2 kHz with a step of 4 Hz. The sound signals were sent and recorded by a vector network analyzer (model: Keysight 5061B). By point-by-point scanning the sound fields in the x-z plane, we can obtain the 2D sound pressure distributions, and the dispersions of Landau levels were obtained by performing the Fourier transformation.

Section 11. Field distributions at A sublattices in the y-z plane

We also place the sound source at the A sublattice and measured the sound pressure distributions at the y-z plane. We depict the measured transition spectrum as the red line in Fig. S8(a), whose minimum is at the frequency of 8500 Hz. The blue line in Fig. S8(a) represents the measured transition spectrum at the C sublattices, whose peak occurs at the frequency of 8500 Hz. The measured sound pressure distributions at A and C sublattices in the y-z plane are shown in left panels of Fig. S8(b, c), which agree well with the simulated counterparts shown in right panels, respectively. As we can see, the propagating gapless zeroth modes can be detected in the C sublattices instead of A sublattices.

Section 12. Comparison of our work with quantum spin/valley Hall effect

Quantum spin Hall effect (QSHE) is a time-reversal invariant system. Due to the spin-orbit coupling, spin-up and spin-down feel opposite synthesized magnetic field, introducing spin-polarized unidirectional currents, as shown in Fig. S9(a). However, our work is more like the valley Hall effect (VHE), where K and K' valleys act as time-reversal counterparts, analogous to spin-up and spin-down states. The synthesized magnetic field is reversed at the K and K' valleys, which introduces valley-polarized unidirectional currents, as shown in Fig. S9(b).

In our work, the K and K' valleys are the time-reversal counterparts to each other, hosting Berry dipoles respectively pointing in the positive and negative k_z directions, as shown in Fig. S9(c). In our experiments, we introduced pseudomagnetic fields utilizing inhomogeneous couplings, resulting in opposite pseudomagnetic fields at the K and K' valleys. Interestingly, the angle between the pseudomagnetic field $\vec{B}$ and the Berry dipole moment $\vec{d}$ remains consistent at both valleys. Therefore, for the same sample in our experiments, $\vec{B}$ is parallel (or antiparallel) to $\vec{d}$ at both K and K' valleys.

In addition, the Hamiltonian is protected by chiral symmetry $\{H_{eff}, U\} = 0$, where $U = \text{diag}(1, 1, -1)$. Chiral symmetry ensures that the zeroth Landau levels must appear in pairs and be symmetric to the zero-energy, even though the Berry dipole at each valley breaks down time-reversal symmetry locally. In the sample 1 where $\vec{B}$ parallel to $\vec{d}$, the zeroth Landau levels at K and K' valleys both behave as a pair of gapless zero modes, as shown as dashed and solid lines with the same color in the top panel of Fig. S9(c). At each valley, the upward- and downward-propagating modes are orthogonal to each other and host opposite chirality, analogous to the helical edge modes in QSHE. Therefore, the gapless zeroth Landau levels can be considered as helical zero modes. Conversely, In the sample 2 where $\vec{B}$ antiparallel to $\vec{d}$, the zeroth Landau levels are gapped out and helical zero modes vanish at both valleys, leading to a band gap with no propagation states, as shown in the bottom panel of Fig. S9(c).

Section 13. Simulated dispersion of the helical zero modes with different losses

In this section, we present simulated dispersion of the helical zero modes with different losses. The source is placed at the center of the bottom surface. Theoretically, we should excite and detect two-way propagating helical states if we disregard losses. However, in real-world experiments, the upward-propagating zero modes decay due to friction losses, resulting in a significant weakening of the reflected sound waves for downward-propagating modes and impacting dispersion measurements. In Fig. S10, it is evident that with an increase in the imaginary part of sound velocity (representing the strength of loss), the simulated dispersion of downward-propagating modes weakens until it nearly vanishes. To overcome this challenge, our experimental approach involved placing sound sources on both the bottom and top surfaces to observe and quantify upward-propagating and downward-propagating helical zero modes, respectively.

To prevent misinterpretation of the two-way propagating helical zero modes as one-way propagating chiral zero modes, we present the measured dispersions of upward-propagating and downward-propagating helical zero modes together in one single figure, as shown in Fig. 4(a). In contrast, Fig. 4(b) shows the measured dispersion spectrum for the sample 2 in the case of $\vec{B} \perp \vec{d}$.

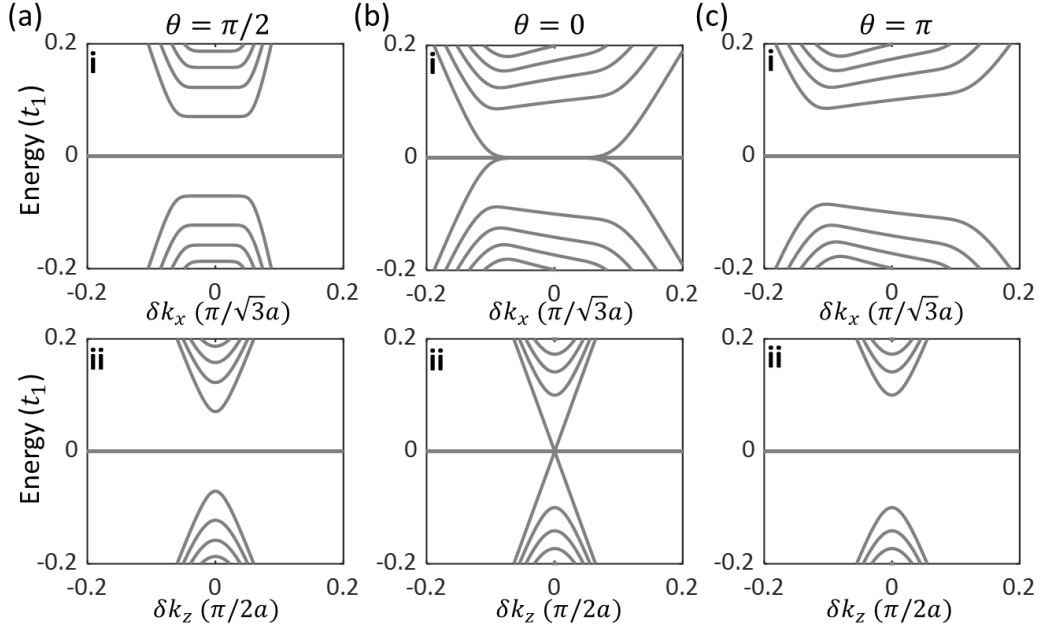


Fig. S1. Landau levels for different cases. (a-c) (i) The spectra of Landau levels in the k_x direction with $k_y = 0$ in the cases of $\theta = \pi/2, 0, \pi$, respectively. (ii) The spectra of Landau levels in the k_y direction with $k_x = -4\pi/(3\sqrt{3}a)$ in the cases of $\theta = \pi/2, 0, \pi$, respectively.

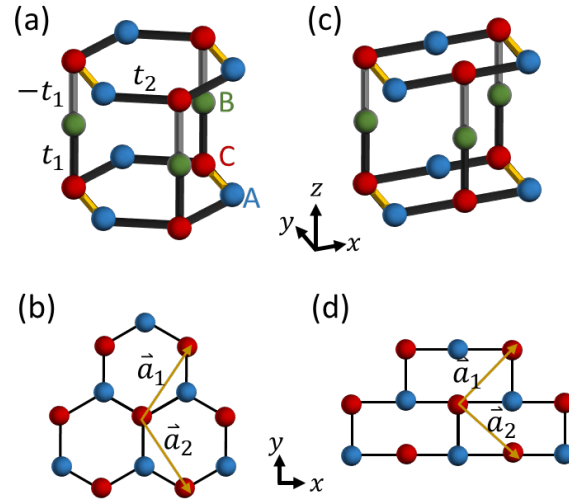


Fig. S2. Hexagonal and tetragonal-like lattice models. (a, b) The tight-binding model of hexagonal and tetragonal-like lattices, respectively. (b, d) The lattice constant vectors of hexagonal and tetragonal-like lattices in the x-y plane, respectively.

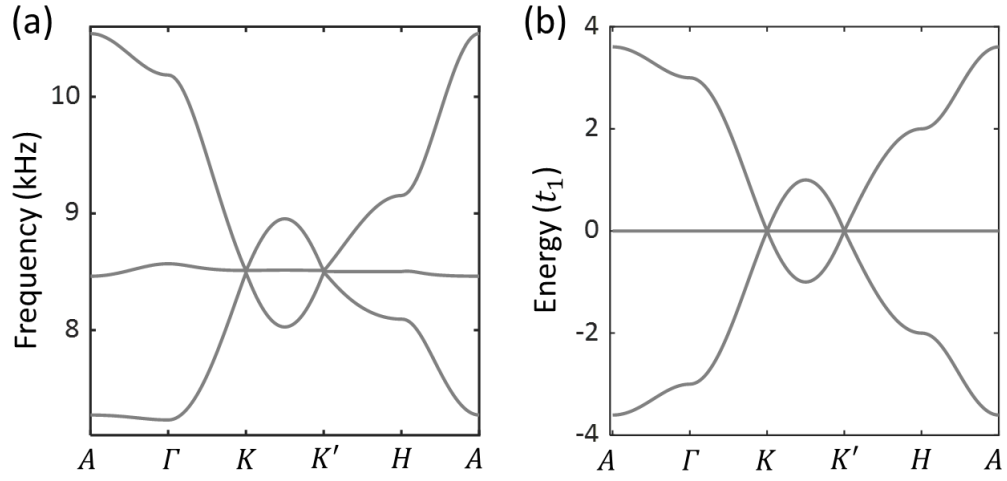


Fig. S3. (a) The simulated dispersion spectrum of the acoustic Berry dipole semimetal. (b) The numerical band spectrum calculated by the tight-binding model. The spectrum in (a, b) is similar, and has almost the same shape around K and K' .

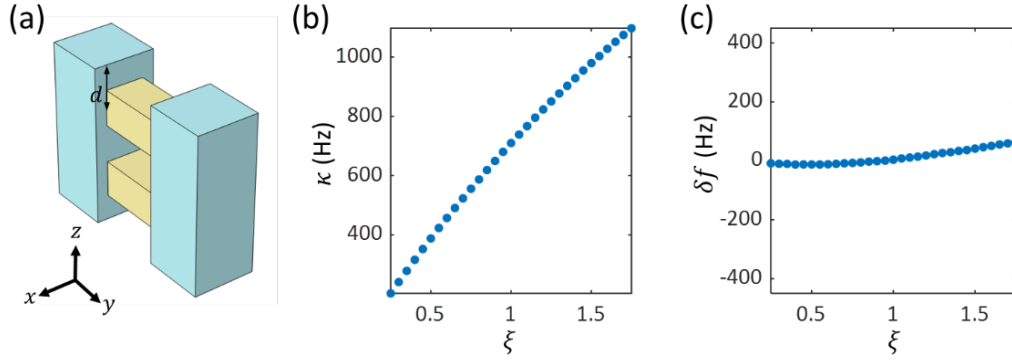


Fig. S4. The coupling strength between two acoustic resonators. (a) Two acoustic resonators coupled with each other. The yellow tubes indicate coupling channels with the location parameter $d = 0.46 \text{ mm}$. (b) The coupling strength against the relative area of cross section of coupling channel ξ . (c) The variation of the center frequency against ξ .

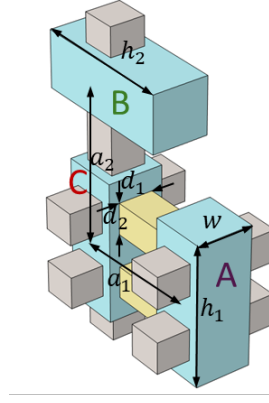


Fig. S5. The size of A and C resonators is $w \times w \times h_1 = 8 \text{ mm} \times 8 \text{ mm} \times 20 \text{ mm}$, while the height of B resonator is slightly different $h_2 = 20.7 \text{ mm}$. The size of cross section of gray coupling channels connecting A and C (B and C) resonators is $d_1 \times d_2 = 4.5 \text{ mm} \times 5 \text{ mm}$ ($d_2 \times d_2 = 5 \text{ mm} \times 5 \text{ mm}$). The size of cross section of yellow coupling channels is controlled by the dimensionless parameter ξ . The width and height of the cross section are $d_1 = \sqrt{1 + \xi} 4.5 \text{ mm}$ and $d_2 = \sqrt{1 + \xi} 5 \text{ mm}$, respectively. The nearest-neighbor distance between A and C (B and C) resonators is $a_1 = 17.6 \text{ mm}$ ($a_2 = 22.2 \text{ mm}$). The size of the whole acoustic samples is $328.8 \text{ mm} \times 297.5 \text{ mm} \times 556.8 \text{ mm}$.

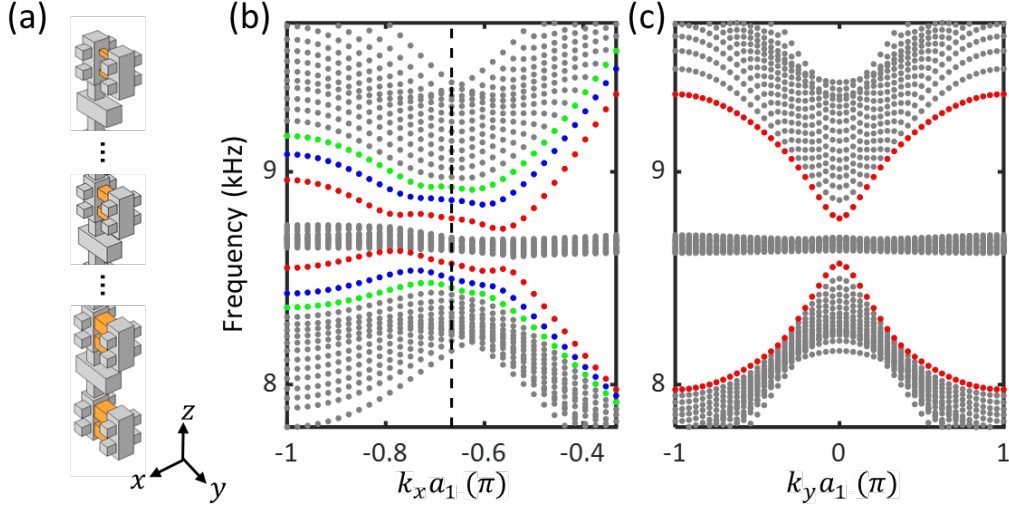


Fig. S6. The simulated Landau levels in the case of $\theta = \pi/2$. (a) Inhomogeneous acoustic superlattices. The pseudomagnetic field is in the y direction with the strength of $0.067a_1^{-2}$. (b) Simulated dispersion spectrum in the k_x direction with $k_y = 0$. The red, blue and green lines denote zeroth, first and second Landau levels, respectively. (c) Simulated dispersion spectrum in the k_y direction the K valley. Red lines denote gapped zeroth Landau levels.

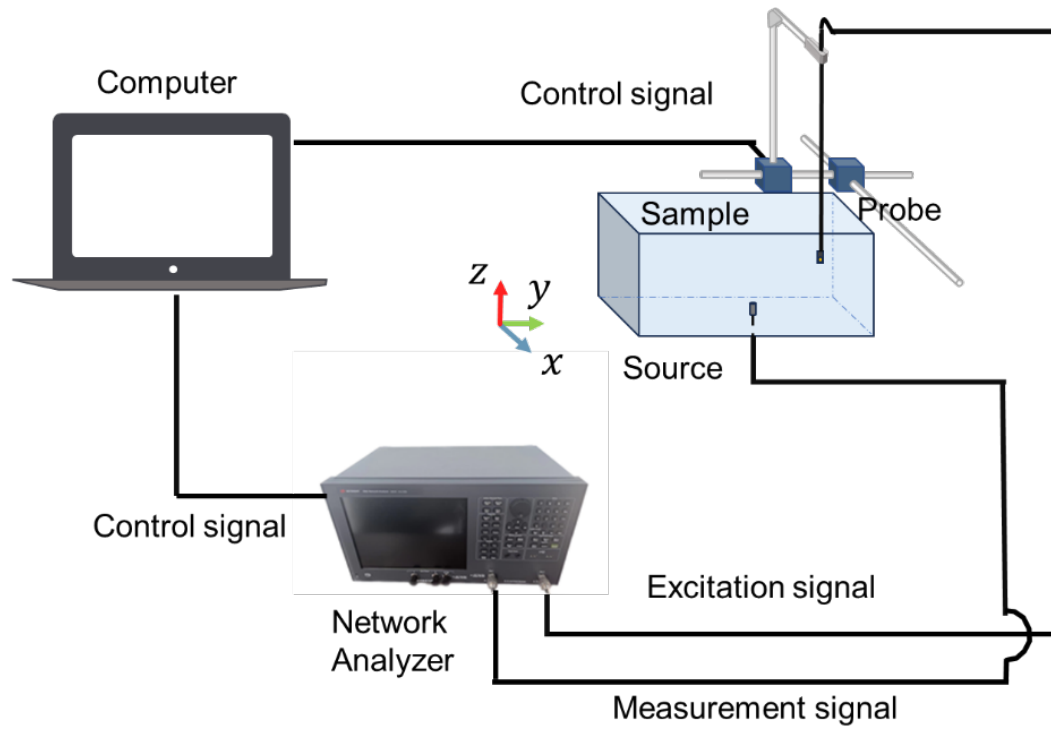


Fig. S7. Experimental setup. The speaker is put at the bottom or the top surface. The sound pressure distributions are measured by inserting a microphone in the sample. Source signal and received signal are generated and recorded by the network analyzer.

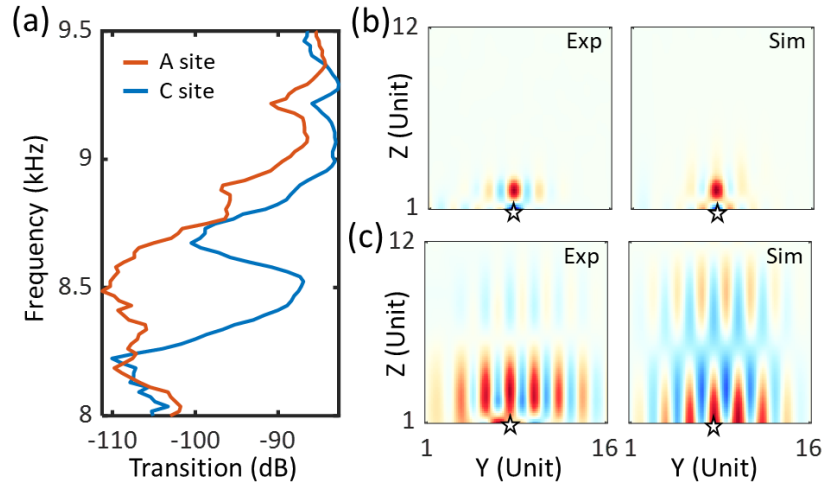


Fig. S8. Experimental detection of field distribution in the A sublattices. (a) Measured transition spectra in A and C sites (red and blue lines). (b) Measured and simulated the sound pressure distribution in the A sites. (c) Measured and simulated the sound pressure distribution in the C sites.

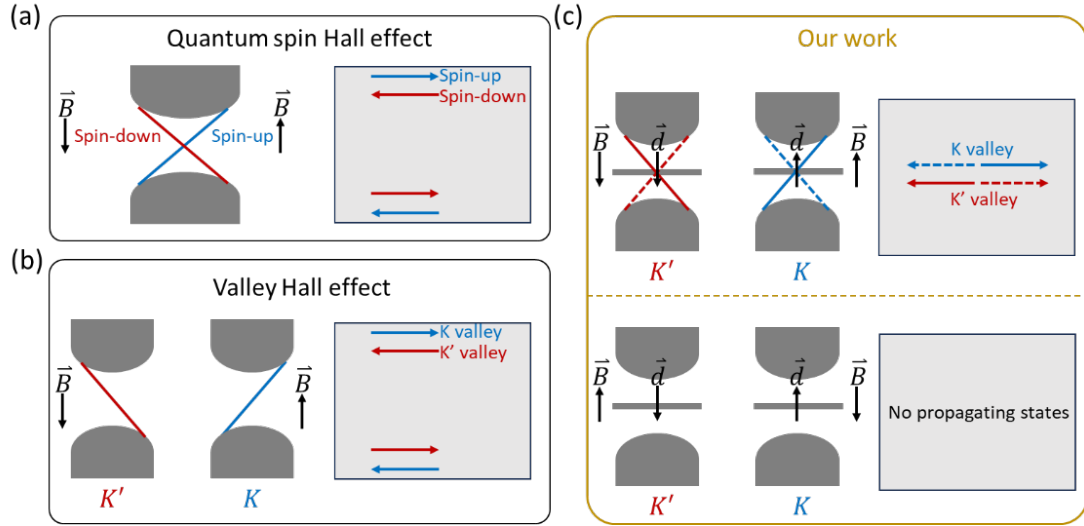


Fig. S9. (a) The dispersion spectrum for the quantum spin Hall effect and the spin-polarized chiral currents. (b) The dispersion spectrum for the valley Hall effect and the valley-polarized chiral currents. (c) The dispersion spectrum for the valley Berry dipole semimetals when $\vec{B} \uparrow \vec{d}$ (top panel) and $\vec{B} \uparrow \downarrow \vec{d}$ (bottom panel). The helical zero modes exist when $\vec{B} \uparrow \vec{d}$ and disappear when $\vec{B} \uparrow \downarrow \vec{d}$.

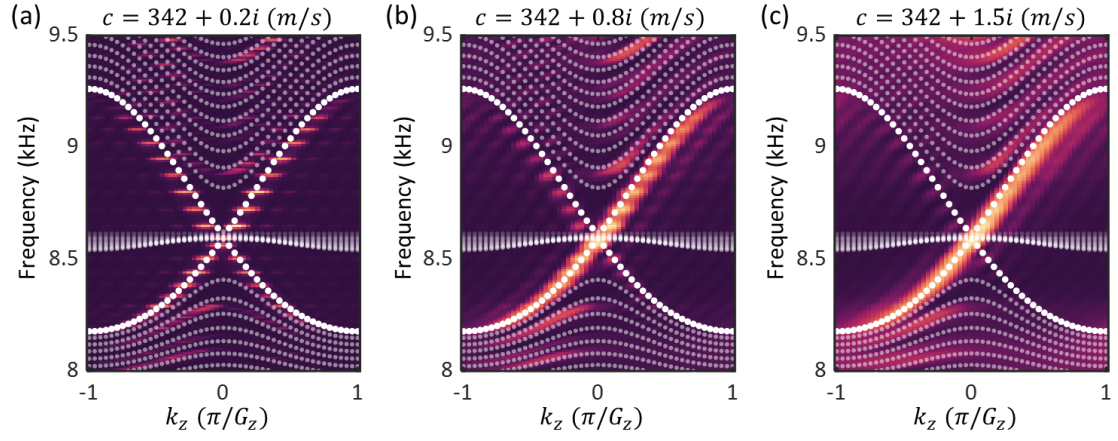


Fig. S10. The colored plots denote simulated dispersion of the helical zero modes when the sound source placed at the bottom surface with the sound velocity $c = 342 + 0.2i$ (a), $c = 342 + 0.8i$ (b), $c = 342 + 1.5i$ (c). The white and gray dots denote the eigenfrequencies of helical zero modes and higher-order Landau level modes, respectively.

Position	d_1 (mm)	d_2 (mm)	ξ
$-7.5a_1$	2.25	2.50	-0.75
$-6.5a_1$	2.66	2.96	-0.65
$-5.5a_1$	3.02	3.35	-0.55
$-4.5a_1$	3.34	3.71	-0.45
$-3.5a_1$	3.63	4.03	-0.35
$-2.5a_1$	3.90	4.33	-0.25
$-1.5a_1$	4.15	4.61	-0.15
$-0.5a_1$	4.39	4.87	-0.05
$0.5a_1$	4.61	5.12	0.05
$1.5a_1$	4.83	5.36	0.15
$2.5a_1$	5.03	5.59	0.25
$3.5a_1$	5.23	5.81	0.35
$4.5a_1$	5.42	6.02	0.45
$5.5a_1$	5.60	6.22	0.55
$6.5a_1$	5.78	6.42	0.65
$7.5a_1$	5.95	6.61	0.75

Table. S1. Structure parameters of inhomogeneous coupling channels in Sample 1. d_1 and d_2 denote the width and length of the cross-section of inhomogeneous coupling channels. ξ is a dimensionless parameter representing the relative area of the cross-section of inhomogeneous coupling channels.

Position	d_1 (mm)	d_2 (mm)	ξ
$-7.5a_1$	5.95	6.61	0.75
$-6.5a_1$	5.78	6.42	0.65
$-5.5a_1$	5.60	6.22	0.55
$-4.5a_1$	5.42	6.02	0.45
$-3.5a_1$	5.23	5.81	0.35
$-2.5a_1$	5.03	5.59	0.25
$-1.5a_1$	4.83	5.36	0.15
$-0.5a_1$	4.61	5.12	0.05
$0.5a_1$	4.39	4.87	-0.05
$1.5a_1$	4.15	4.61	-0.15
$2.5a_1$	3.90	4.33	-0.25
$3.5a_1$	3.63	4.03	-0.35
$4.5a_1$	3.34	3.71	-0.45
$5.5a_1$	3.02	3.35	-0.55
$6.5a_1$	2.66	2.96	-0.65
$7.5a_1$	2.25	2.50	-0.75

Table. S2. Structure parameters of inhomogeneous coupling channels in Sample 2. d_1 and d_2 denote the width and length of the cross-section of inhomogeneous coupling channels. ξ is a dimensionless parameter representing the relative area of the cross-section of inhomogeneous coupling channels.

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